

Vineet Agarwal

240-353-9811 — vineetagarwal540@gmail.com — vineetagarwal54 — Github — Portfolio

Summary — Software engineering graduate student with 2+ years of production experience building full-stack applications and cloud-native backends. Designs and ships REST APIs, serverless AWS pipelines, and containerized services on Kubernetes, with a track record of cutting latency and defects in live systems. Comfortable across the stack from React front ends to the data layer, pursuing a Cloud Engineering focus.

Skills

Languages Python, JavaScript, TypeScript, C++, SQL, Go
Backend FastAPI, Node.js, Express, REST APIs, GraphQL, WebSocket, Socket.IO
Cloud and DevOps AWS (Lambda, API Gateway, EKS, S3, DynamoDB, SQS, CloudFront, CloudFormation), Docker, Kubernetes, CI/CD (GitHub Actions), Azure Functions

Frontend React, Next.js, React Native, Redux Toolkit
Databases PostgreSQL, MongoDB, Redis, FAISS, Pinecone
Applied AI multi-agent systems, RAG, LangChain, OpenAI APIs, Claude, prompt engineering, embeddings, vector search
Architecture and Practices System Design, Microservices, Serverless, Auth (JWT, RBAC, SSO), Agile

Experience

ServBeyond Solutions

Jun 2026 – Present

Enterprise AI & Platform Intern

- Engineered platform integrations across Salesforce, ServiceNow, and AWS, wiring REST and SOAP services into a normalized data layer that powers company-wide automation for 100 employees.
- Streamlined meeting-to-ticket and opportunity intake with Python Azure Functions and Power Automate, deployed company-wide and saving up to 2 hours per meeting and 10 to 15 minutes per opportunity.
- Shipped a retrieval assistant over a version-controlled SharePoint archive used daily by 20+ staff, pairing auto-vectorized ingestion with a React front end.

Runara.ai

March 2026 – May 2026

Software Engineer Intern

- Implemented speculative decoding from scratch in PyTorch and HuggingFace across two GPUs, applying the rejection-sampling acceptance criterion and establishing the draft-to-target cost ratio under which the technique yields 2 to 3x inference throughput on large target models.

Svipes

Jul 2024 – Dec 2024

Software Engineer

- Optimized a video module with async loading and client-side caching, cutting playback stutter 70% and load time by 1.3 seconds across iOS and Android.

Xelpmoc Design and Tech Limited

Nov 2022 – Apr 2024

Software Engineer

- Reduced API response time from 75s to under 10s and raised throughput 4x by refactoring SQL joins, indexing high-traffic tables, and adding a Redis caching layer.
- Architected RESTful backend services for a tourism platform, cutting production defects 40% and accelerating feature delivery 25%.
- Designed a real-time interview system with WebRTC and Socket.IO supporting 100+ concurrent sessions at sub-200ms latency.

IIIT Hyderabad

Jun 2022 – Aug 2022

Software Intern

- Automated multilingual data extraction across 5+ regional languages with Python and Selenium, improving NLP classification accuracy 15%.

Education

University of Maryland, College Park

Jan 2025 – Dec 2026 (Expected)

Master of Engineering, Computer Software Engineering — Minor in Cloud Engineering

GPA: 3.7

Projects

Serverless Video Analytics Platform, AWS (Github)

- Deployed a serverless video pipeline on AWS Lambda, API Gateway, DynamoDB, and SQS with CloudFormation and GitHub Actions, using dedicated upload, processing, and analytics functions auto-scaling behind CloudFront to absorb bursty load at low idle cost.

Locra: On-Device Multimodal AI, Android (Github)

- Developed an Android-first multimodal AI assistant in React Native, TypeScript, and Expo that runs a quantized Qwen3-VL-2B vision-language model fully on device via llama.rn, pairing on-device Whisper speech-to-text with a Kotlin native module that enforces model integrity and a hard offline guarantee.